



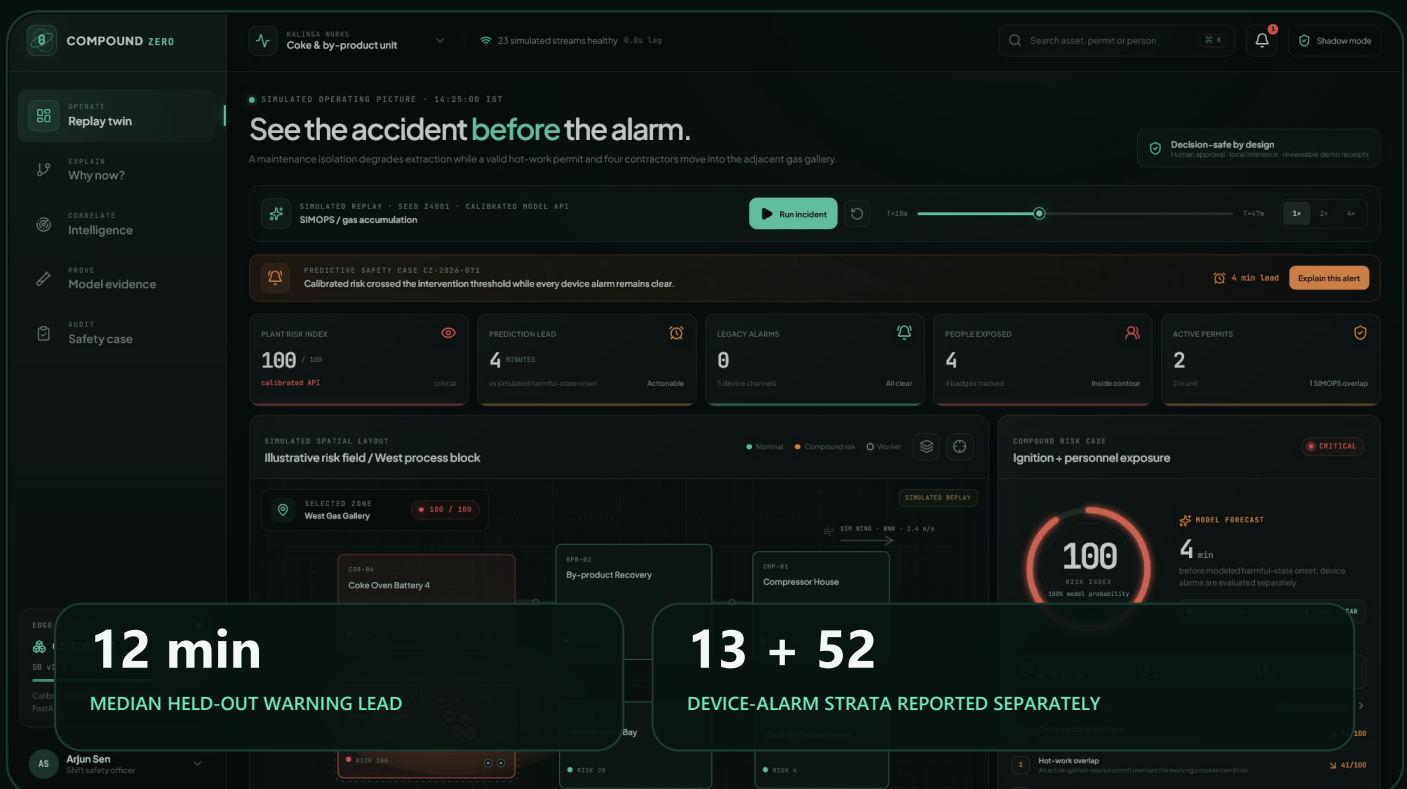
See the accident before the alarm.

A human-gated industrial safety intelligence prototype that turns weak process, permit, barrier and personnel signals into one reviewable risk case.

NO LLM IN RISK PATH

MANUAL ONLY

SIMULATED EVIDENCE



Detailed submission | Prepared by Parth Katiyar | 22 July 2026

SIMULATED DATA / PROTOTYPE / NOT FIELD VALIDATION / NOT AUTONOMOUS CONTROL



EXECUTIVE THESIS

One risk case. Five witnesses.

Compound Zero wins on disciplined evidence: an early-warning story judges can see, metrics they can reproduce, and safety limits the product cannot cross.

THE BET

The blind spot is not missing data.
It is missing interaction intelligence.

A gas trend below its device threshold can become urgent when extraction is impaired, hot work is active and people enter the same zone. Compound Zero detects that combination before treating any one sensor as decisive.

01

Inspectability

Fixed-seed replay, typed factors, raw traces, dataset and model hashes.

02

Decision safety

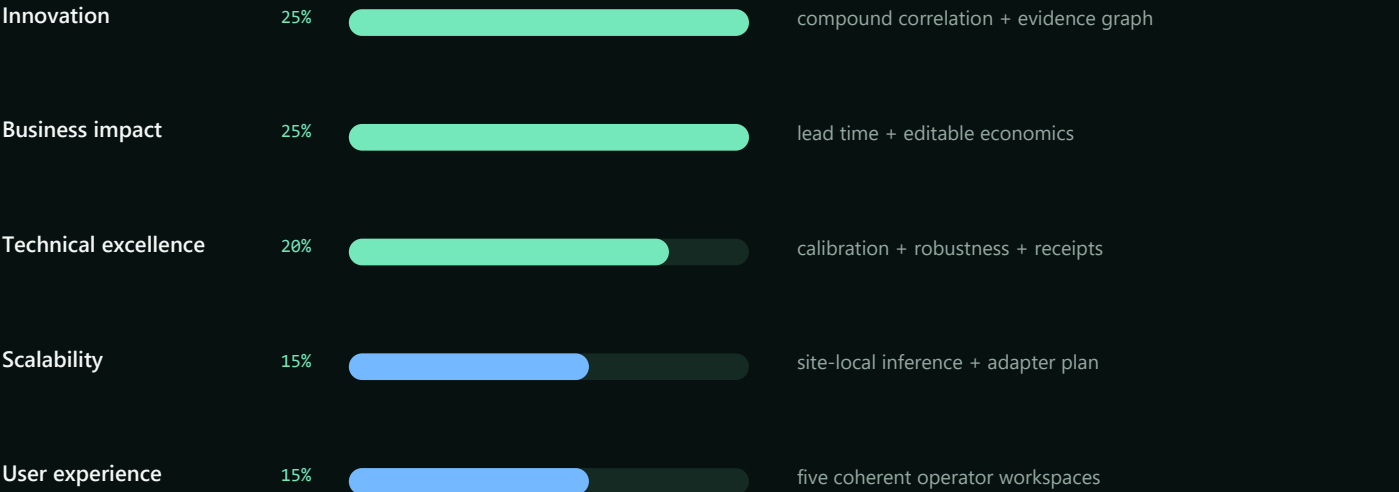
Advisory output, stream-quality abstention and no actuator route.

03

Transfer honesty

LOSO exposes a 0% process-drift fold and turns it into a pilot gate.

JUDGING FIT





PROBLEM STATEMENT FIT

The brief, translated into evidence.

The prototype addresses the central compound-risk challenge while keeping every unimplemented connector and production control visible.

BRIEF CAPABILITY	IMPLEMENTED EVIDENCE	DECLARED BOUNDARY
IoT + SCADA	Typed signals, trends, quality and fixed-seed replay	Live historian adapter
Permit to work	Permit context, evidence binding and conflict checks	Authorized PTW connector
CCTV / location	Anonymous event metadata; zone count and distance	No raw video or biometrics
Shift + maintenance	Handover and extraction-barrier features	CMMS / workforce adapters
Compound detection	Calibrated 28-feature full-fusion classifier	Site calibration
Pre-emptive response	Human-gated dry-run plan and audit receipt	No field actuation
Geospatial layer	Fictional plant polygons + uncertainty review	Surveyed site validation
Incident intelligence	Local cited BM25 over 5 authored patterns	No full regulatory RAG

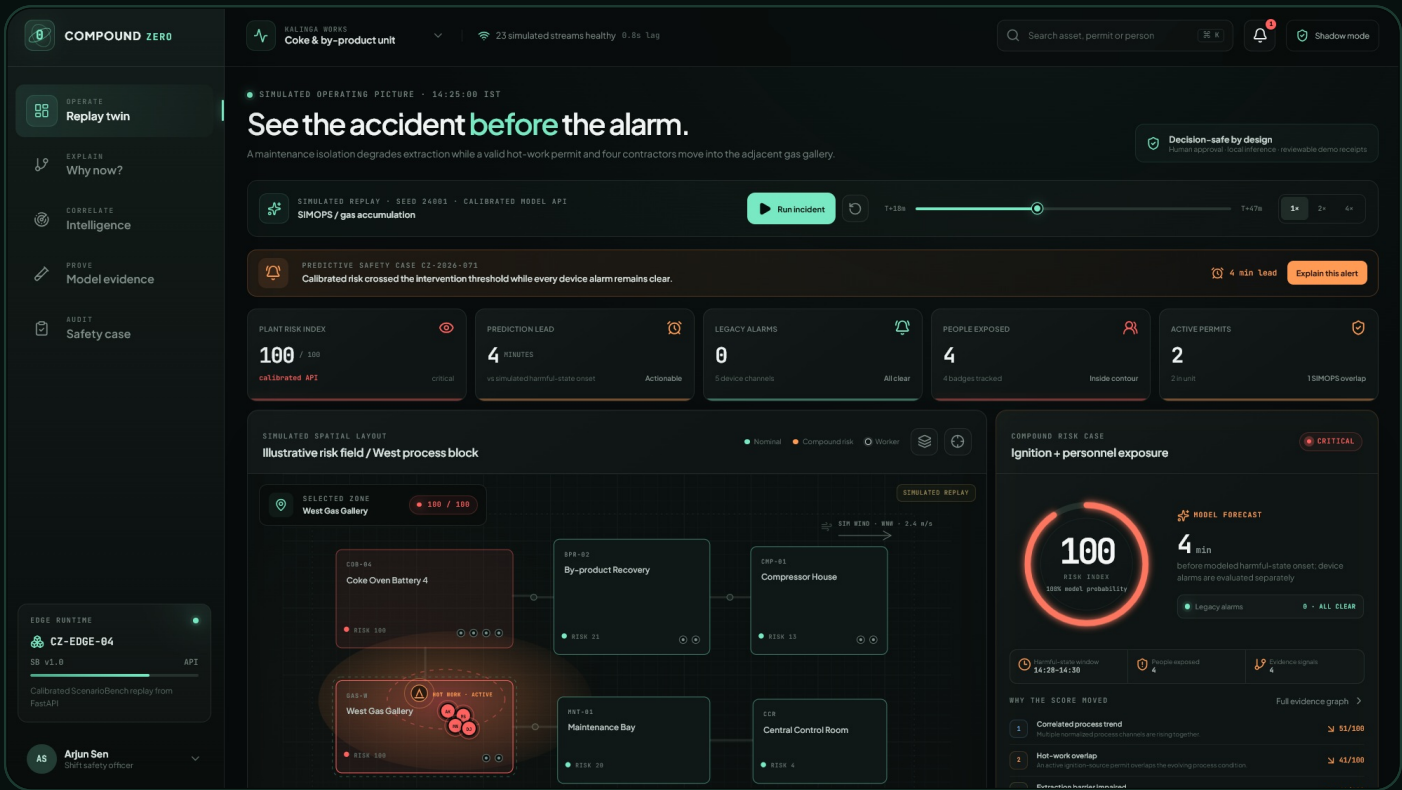
WHY THIS IS CREDIBLE

- The same persisted model powers API replay and the measured benchmark.
- Device-alarm groups and no-device-alarm groups use separate denominators.
- The safety case says what is absent: field connectors, identity, durable audit and actuation.

PRODUCT EXPERIENCE

Five workspaces. One safety case.

The interface moves from operating picture to explanation, cross-system evidence, model proof and an exportable safety record.



- 1

Replay twin

See risk, people, permits and lead
- 2

Why now?

Inspect typed evidence and traces
- 3

Intelligence

Join vision, permits and patterns
- 4

Model evidence

Compare baselines and limits
- 5

Safety case

Record authority and evidence

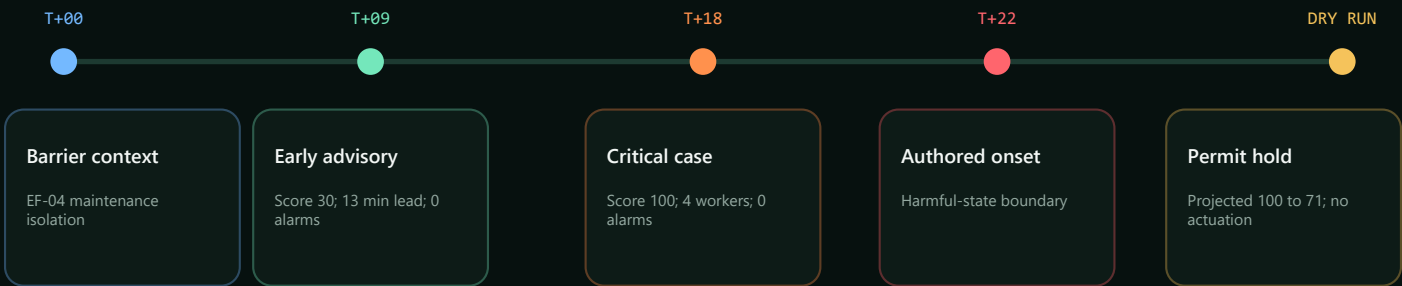
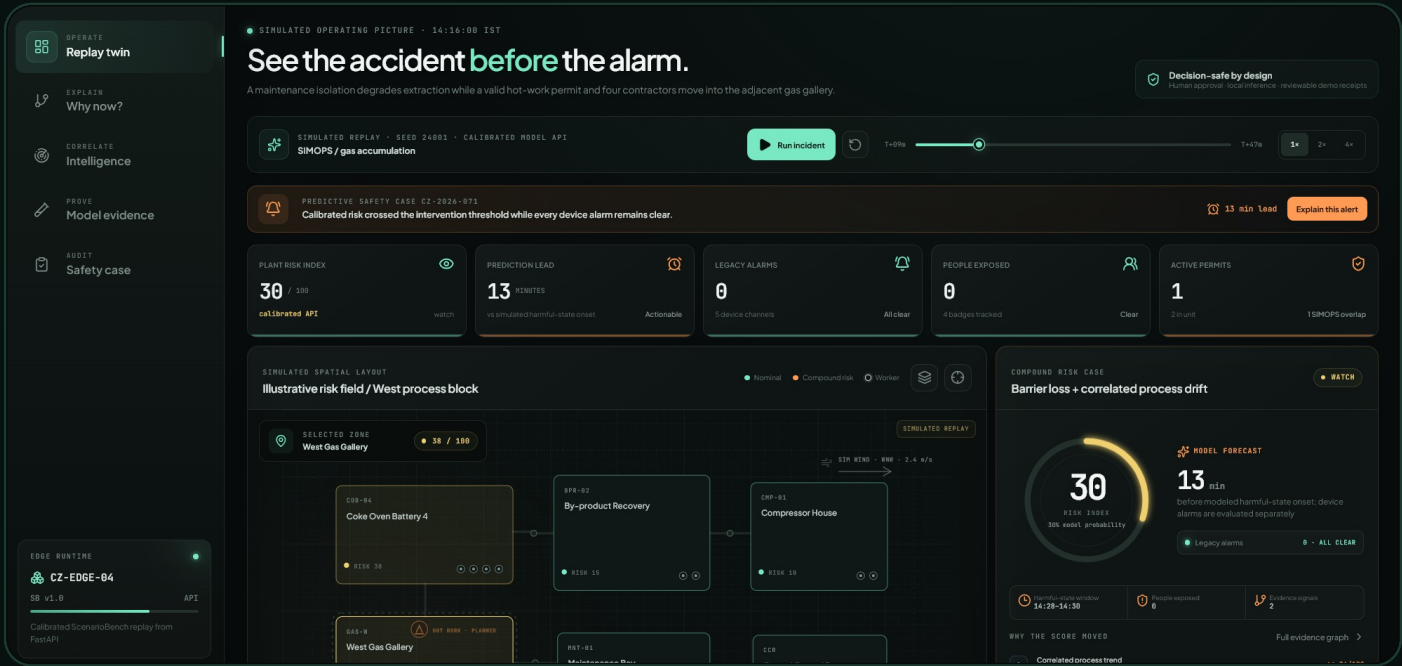
DESIGN PRINCIPLE

Explain evidence first. Propose controls second. Keep authority human.

SIGNATURE REPLAY

A warning while every device alarm is clear.

Seed 24001 is a deterministic demonstration, not a reconstruction of a real incident. Harmful-state timing is authored for evaluation.



COUNTERFACTUAL BOUNDARY

The permit-hold number is a deterministic scenario projection. It is not a causal estimate, does not preserve the original model probability or lead-time truth, and executes no plant action. The operator records only a dry-run decision.



TECHNICAL ARCHITECTURE

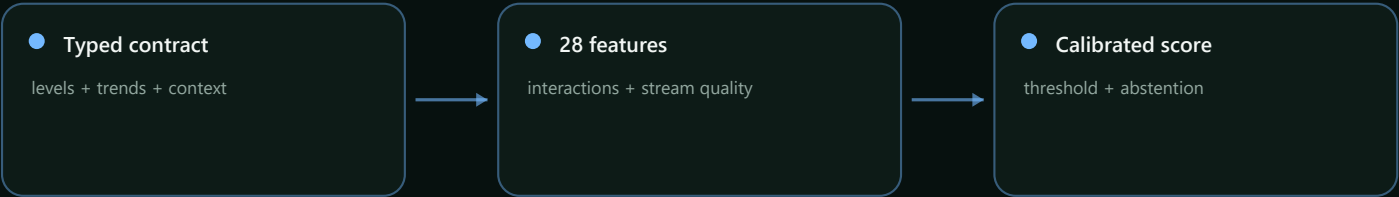
Numeric risk path, evidence path, authority path.

The model can fail without disabling physical alarms or silently crossing into plant control. Explanation is downstream of the score.

OFFLINE EVIDENCE



ONLINE RISK



HUMAN AUTHORITY



IMPLEMENTED NOW

- FastAPI persisted-model replay
- React five-workspace command center
- Local BM25 + deterministic audit

HARD BOUNDARY

- No LLM in scoring or policy
- No actuator client or route
- Physical alarms and SIS stay independent



MODEL AND DATA

Reproducible by construction.

The prototype uses calibrated gradient boosting over deterministic simulated replays. No generator or language model participates in the decision path.

27,648

ROWS

9 scenario families

576

REPLAYS

64 deterministic seeds

5,616

TEST ROWS

13 held-out seeds

0

SPLIT OVERLAP

train / calibrate / test

FEATURE CONTRACT

Process

LEL, H2S, CO, oxygen, pressure; levels and slopes

Barriers

ventilation impairment, isolation and stream quality

Work context

hot work, confined space, handover and overlaps

Exposure

worker count, proximity and anonymous event confidence

Interactions

gas x hot work, gas x worker, barrier x gas and more

MODEL RECEIPT

FIELD	RECORDED VALUE
Engine	Histogram gradient boosting + sigmoid calibration
Decision threshold	0.155, selected on calibration only
Dataset SHA-256	96166a2762577a2fa8440e2a6f0f819c342aef9e9cf790e3ed2bef3eb6f4e314
Model SHA-256	ad8499f7afc5d6344d1f9a1b6e4fee200f44dc8cf8e77e652d5b204b1f047e84
Artifact bytes	238,184

Joblib artifacts are unsafe if untrusted. The API verifies the expected SHA-256 before deserialization; production still requires a signed registry and restricted write path.



HELD-OUT EVIDENCE

Context closes the blind spot.

The test split contains 5,616 rows and 65 event replays. Results are reproducible on authored simulated scenarios and are not a field-performance claim.

METHOD	EVENT RECALL	EVENT FNR	LEAD	FA / 24 H	ROW AP	BRIER
Single sensor	20.0%	80.0%	5 min	7.50	0.813	0.242
Process only	100.0%	0.0%	12 min	8.65	0.947	0.070
Full fusion	100.0%	0.0%	12 min	1.15	0.997	0.015

EVENT RECALL



FALSE-NEGATIVE DELTA

-80 pp

Full fusion versus device baseline at event level.

FALSE-ALARM DELTA

-86.7%

Full fusion versus process-only episodes per simulated 24 h.

Interpretation: ablation supports the compound-risk hypothesis inside ScenarioBench. The result does not establish site transfer, operational utility, safe alert thresholds or real incident prevention.



BASELINE COMPARISON

Never fold 'no device alarm' into 'earlier'.

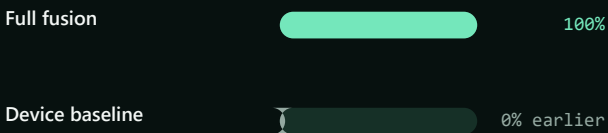
The comparison window ends at each authored harmful-state onset. Two different questions therefore use two different denominators.

STRATUM A

13 / 13

DETECTED BEFORE DEVICE ALARM

Only event replays where a device alarm exists by harmful-state onset belong in this denominator.



STRATUM B

52 / 52

DETECTED WITH NO DEVICE ALARM

These event replays have no device alarm by onset. They answer detection, not earlier-than-device timing.



WHY THIS MATTERS

A folded rate would overstate the comparison by calling an absent device alarm a timing win. Compound Zero reports the 13 comparable groups and 52 no-alarm groups separately, then shows the device baseline's 13 / 65 event recall.

DENOMINATOR RECEIPT

65 authored event groups = 13 with a device alarm by onset + 52 without one. Full fusion detects 65 / 65; the single-device rule detects 13 / 65. These are simulated group-level results.



ROBUSTNESS AND LIMITS

The failure we found matters.

A frozen-model stress suite and true scenario-type holdout reveal where the strong base split does not transfer.

100%

BASE EVENT RECALL

65 held-out simulated events

0.006

BASE ECE

10 equal-width probability bins

80%

LOSO MACRO RECALL

5 event-bearing scenario folds

LEAVE-ONE-SCENARIO-TYPE-OUT

HELD-OUT SCENARIO TYPE	EVENT RECALL	TRAINING RELATION
barrier loss workers	100%	Unseen scenario family
compound confined space	100%	Unseen scenario family
compound handover	100%	Unseen scenario family
compound hot work	100%	Unseen scenario family
process drift	0%	Unseen scenario family

FAILURE: PROCESS_DRIFT = 0%

The model misses the event-bearing process_drift family when that entire family is absent from training. This blocks any transfer claim and makes unseen operating modes, local drift and shadow-mode replay mandatory pilot gates.

FROZEN-MODEL STRESS

STRESS	RECALL	MEDIAN LEAD	FA / 24 H
Baseline	100%	12 min	1.15
Barrier + shift missing	80%	7 min	1.15
Process stream missing	80%	12 min	7.50
20% packet dropout	100%	12 min	39.23
Sensor noise sigma 0.05	100%	12 min	47.88

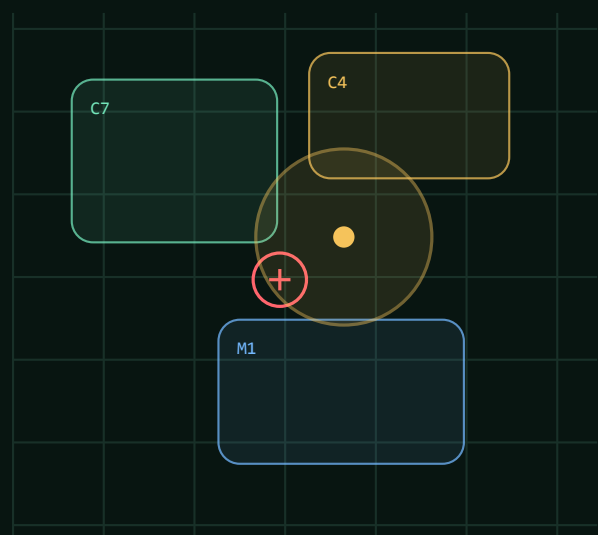


GEOSPATIAL EVIDENCE

Uncertainty becomes review, not false precision.

GeoEvidenceBench tests fictional site-local polygons and nearest authored hazards on a boundary-heavy software-conformance fixture.

FICTIONAL PLANT-LOCAL METRES



Declared error bound routes boundary overlap to human review.

337

SIMULATED CASES
141 positive / 196 negative

0.907 m

DISTANCE MAE
p95 absolute error 2.255 m

1.000

CERTAIN-EXPOSURE PRECISION
on this authored fixture

CLASSIFICATION TRADE-OFF

METHOD	RECALL	FNR	FPR	POLICY
Point estimate	0.816	0.184	0.168	Decide from reported point
Uncertainty review	1.000	0.000	0.454	Route possible overlap

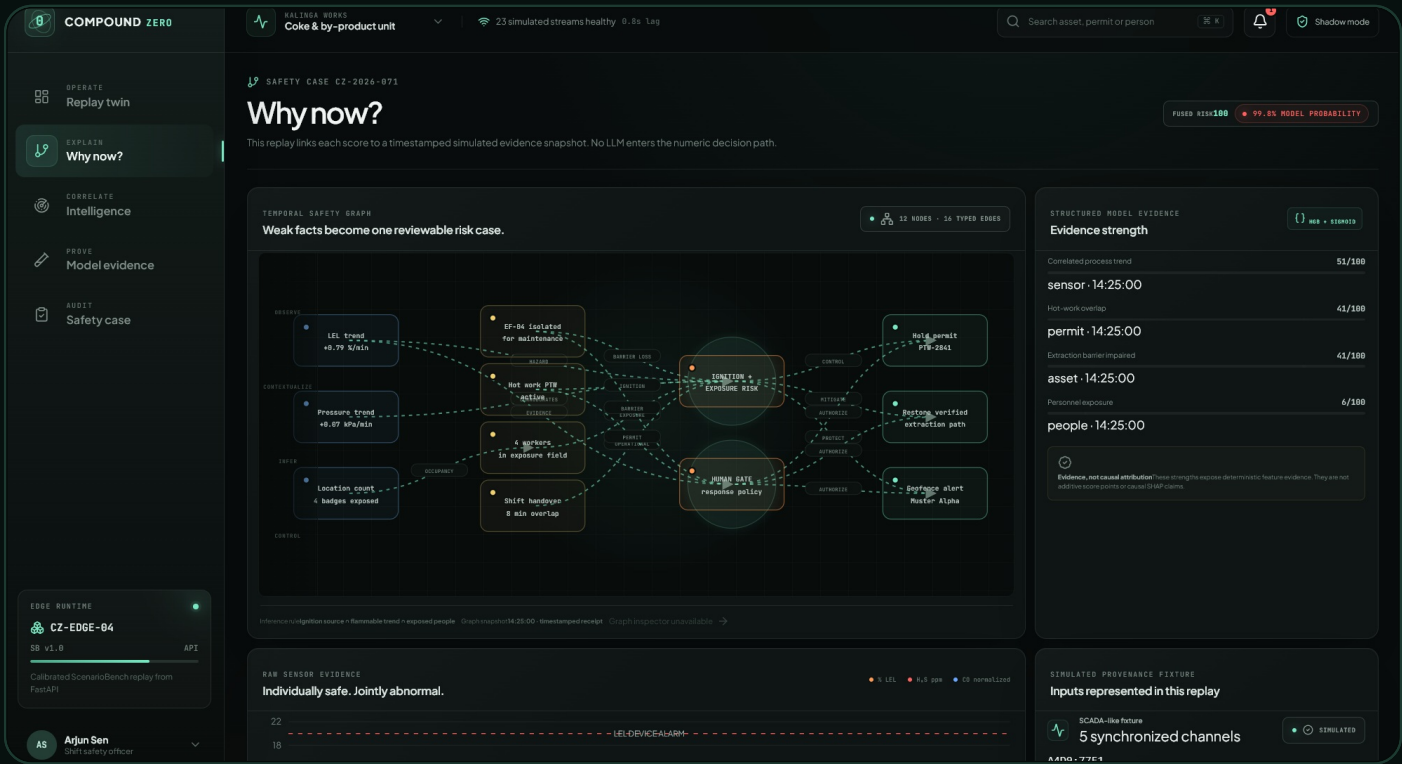
BOUNDARY

Review recall rises from 0.816 to 1.000 while false-positive review rate rises from 0.168 to 0.454. That is workload, not hidden accuracy. No surveyed coordinates, plume physics, representative prevalence or field geospatial performance is claimed.

EVIDENCE EXPERIENCE

Weak facts become one reviewable case.

Structured factors, timestamps and typed edges give an operator a path from score to source without pretending feature contribution is causality.



WHAT IS SHOWN

Sensor, permit, asset, people and policy evidence with typed relationships and source timestamps.

WHAT IS NOT SHOWN

No causal SHAP claim, opaque narrative, legal conclusion or language-model decision.

CROSS-SYSTEM INTELLIGENCE

Observe

Anonymous CCTV event metadata only

Correlate

Numeric fusion, no raw frames

Interrogate

Local cited BM25, no generator

Authorize

Role-gated manual response



SAFETY AND AUTHORITY

Machine proposes. People authorize.

Consequence determines the role gate. Every prototype action remains a record or browser dry run; no equipment, access-control or emergency system is invoked.



AUTHORITY LADDER

ACTION CLASS	REQUIRED AUTHORITY	PROTOTYPE / PRODUCTION BOUNDARY
Advisory display	May notify	No physical-world change
Permit hold	Permit authority	Reversible workflow; prototype dry run
Evacuation	Incident command	Existing emergency procedure
Process isolation	Dual authorization	Existing control system only
Shutdown / interlock	SIS / operations	Not implemented; remains independent

SAFE FAILURE

Stream quality below 0.80 causes abstention and suppresses model prediction. Existing alarms never depend on the AI path.

CURRENT GAP

No SSO, durable signed ledger, TLS, live connector, production freshness policy or tested failover exists yet.



CURRENT-LAW ADVISORY MAP

Navigation aids, never a compliance verdict.

The map uses current central primary sources as of 22 July 2026. Applicability, state rules and sector obligations require a qualified owner with authorized text.

PRIMARY REFERENCE	ADVISORY MAPPING	BOUNDARY
OSH&WC Code s18 + Second Schedule	Safety standards; items 16, 18 and 23 cover dangerous fumes/gases, explosive atmospheres and danger prohibition	Evidence categories only
OSH&WC Code s84	Hazard information, measures for hazardous substances and on-site emergency planning for covered hazardous-process factories	Supports evidence completeness
OSH&WC Code s89	Worker notice of imminent danger and immediate remedial action / referral route	Supports escalation design
First Schedule	Includes integrated steel, coke/fuel-gas and highly flammable gas industries	Applicability remains site-specific
Central Rules, 2026	G.S.R. 345(E), 8 May 2026; effective on Gazette publication	Current central rules reference
OISD-STD-105 catalogue	Public metadata for work permit system	Normative text not bundled

IMPORTANT PTW LIMIT

A text review of the 2026 Central Rules identified no general central rule explicitly labelled 'hot work', 'work permit' or 'permit to work'. This is not a legal conclusion. State rules, site procedures, sector regulations, licence conditions and authorized OISD standards may govern the work.

OSH&WC Code, 2020

Commencement S.O. 5321(E), 21 Nov 2025

OSH&WC Central Rules, 2026 - G.S.R. 345(E)



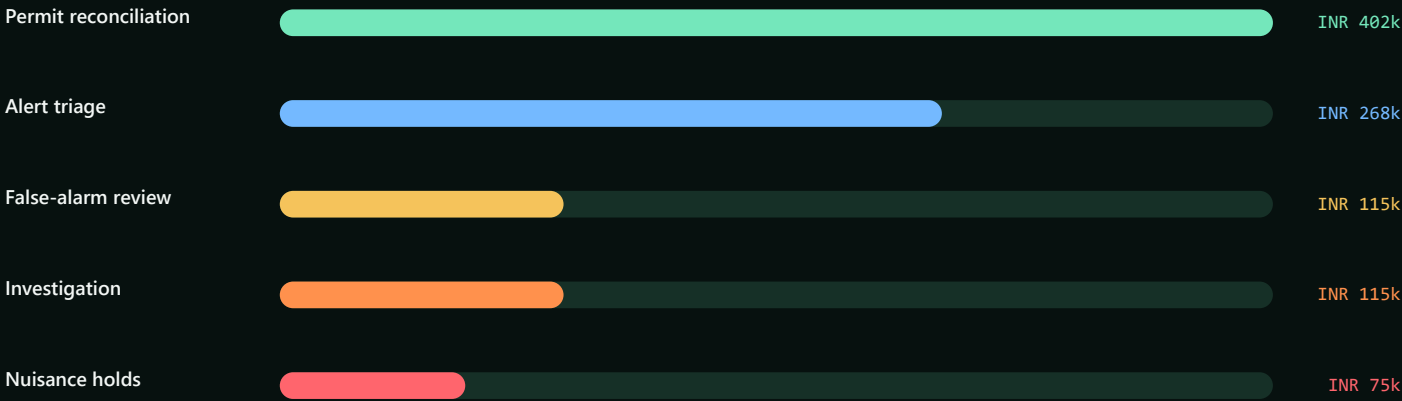
PILOT ECONOMICS

A sensitivity model that is allowed to fail.

The editable low/base/high cases value workflow capacity and nuisance-hold exposure only. They exclude human harm and prevented-incident valuation.

CASE	ANNUAL BENEFIT	NET AFTER OPEX	PAYBACK	3-YEAR BCR
Low	INR 38,008	INR -261,992	Not reached	0.07
Base	INR 973,875	INR 523,875	25.2 mo	1.19
High	INR 3,975,157	INR 3,275,157	6.6 mo	3.06

BASE-CASE COMPONENTS



ETHICAL AND FINANCE BOUNDARY

- No value is assigned to injury, death, catastrophic loss or a prevented incident.
- Benefits are illustrative capacity and nuisance-downtime exposure, not guaranteed cash savings.
- The low case does not pay back; the high case is an upside bound, not a forecast.



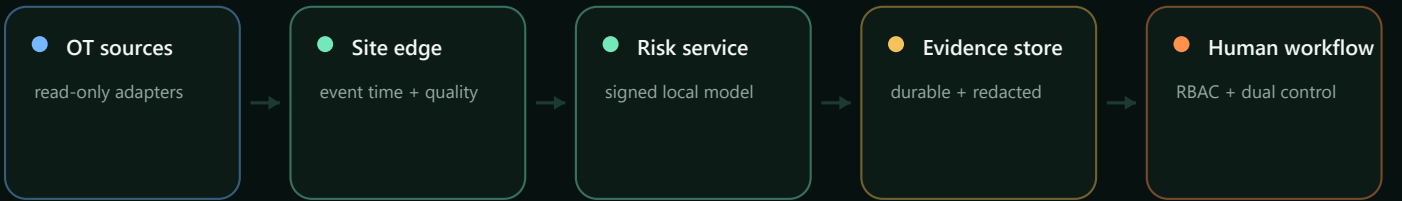
SCALABILITY AND PILOT

Ninety days to earn the right to continue.

The next phase is a read-only shadow pilot with measurable stop conditions, not a shortcut from prototype to production control.



TARGET DEPLOYMENT SHAPE



FIELD GATES

- Process-drift family and local operating modes must pass held-out shadow replay.
- Alarm burden under dropout / noise must stay inside an owner-approved budget.
- AI failure must not alter detector, SIS, permit or emergency authority.



SUBMISSION TRACEABILITY

Every judging criterion has an artifact.

This matrix points judges to inspectable evidence and names the proof still required before any field claim.

CRITERION	IMPLEMENTED EVIDENCE	WHERE TO INSPECT
Innovation 25%	Interaction model, typed evidence graph, uncertainty review, human authority	Replay twin; Why now?; Intelligence
Business impact 25%	12 min simulated median lead; workflow sensitivity model	metrics.json; pilot_economics.json
Technical excellence 20%	Whole-seed split, calibration, model hash, LOSO, stress suite	model_card.json; robustness_metrics.json
Scalability 15%	Stateless inference base, site-local target, read-only adapter plan	architecture.md; Docker / CI
User experience 15%	Five workspaces, deterministic replay, evidence export and dry-run studio	apps/web; demo video

EVALUATION FOCUS

FOCUS	MEASURED / IMPLEMENTED	BOUNDARY
Compound detection	Full fusion recall 1.00 vs device 0.20	Simulated held-out event groups
Lead time	12 min vs device 5 min	Authored harmful-state timing
Geospatial	Review recall 1.00; certain precision 1.00	Boundary-heavy conformance only
Regulatory coverage	Current-law advisory map	No control-by-control assessment
False negatives	0.80 absolute event-FNR reduction	Not site-transfer evidence

Deliverables: working prototype, architecture diagram, presentation deck, detailed document and demo-video production package. The repository is the verification surface; this report is a map, not a substitute for inspection.



CLAIM BOUNDARIES AND EVIDENCE

Strong enough to inspect. Honest enough to trust.

The project distinguishes what exists now, what is simulated, and what a qualified site owner must validate next.

NOT CLAIMED

- Field-validated or certified safety performance.
- Autonomous evacuation, shutdown, isolation, interlock or permit authority.
- Live SCADA, PTW, CMMS, worker-location or raw CCTV integration.
- A complete legal / OISD corpus or a compliance determination.
- A causal intervention estimate, plume model or surveyed plant geometry.
- Guaranteed ROI, prevented incidents or monetary valuation of human harm.

CHECKED-IN EVIDENCE

`artifacts/metrics.json`

`artifacts/robustness_metrics.json`

`artifacts/geospatial_metrics.json`

`artifacts/pilot_economics.json`

`artifacts/model_card.json`

`data/scenario_bench/metadata.json`

`docs/security-safety.md`

`docs/requirements-traceability.md`

REPRODUCE

```
python -m ml.train
python -m ml.robustness_bench --output artifacts/robustness_metrics.json
python -m ml.geospatial_bench --output artifacts/geospatial_metrics.json
python -m ml.pilot_economics --output artifacts/pilot_economics.json
python -m pytest -q | pnpm test | pnpm typecheck | pnpm build
```

PRIMARY SOURCE LEDGER

1. Occupational Safety, Health and Working Conditions Code, 2020

https://labour.gov.in/sites/default/files/osh_gazette.pdf

2. S.O. 5321(E) commencement notification, 21 November 2025

<https://labour.gov.in/sites/default/files/e-noti-osh-1.pdf>

3. OSH&WC (Central) Rules, 2026 - G.S.R. 345(E), 8 May 2026

<https://www.labour.gov.in/static/uploads/2026/05/ee246f790cad0b8e99c3828f34fa09a6.pdf>

4. OISD standards catalogue

<https://www.oisd.gov.in/en-in/oisd-standards-list>

5. ISO 45001 overview

<https://www.iso.org/standard/63787.html>

6. Compound Zero public repository

<https://github.com/Fyxod/compound-zero>

See the accident before the alarm.